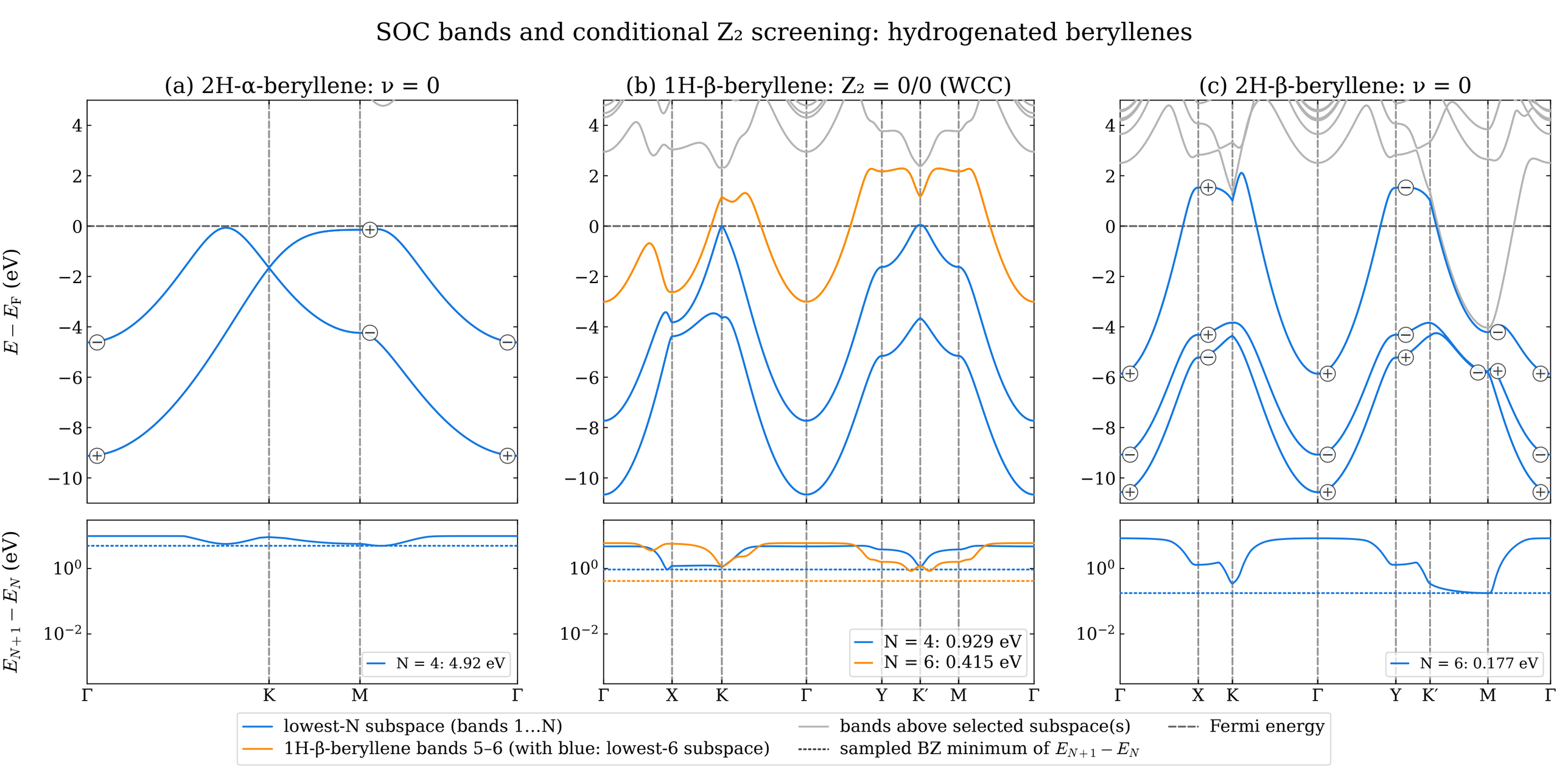

SOC bands and conditional Z₂ screening: hydrogenated beryllenes
(a) 2H-α-beryllene: ν = 0
(b) 1H-β-beryllene: Z₂ = 0/0 (WCC)
(c) 2H-β-beryllene: ν = 0
$E - E_F$ (eV)
$E_{N+1} - E_N$ (eV)
Γ K M Γ
Γ X K Γ Y K′ M Γ
N = 4: 4.92 eV
N = 4: 0.929 eV
N = 6: 0.415 eV
N = 6: 0.177 eV
lowest-N subspace (bands 1…N)
1H-β-beryllene bands 5–6 (with blue: lowest-6 subspace)
bands above selected subspace(s)
sampled BZ minimum of $E_{N+1} - E_N$
Fermi energy